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Review

Effects of Coenzyme Q10 Supplementation on Metabolic Indicators in Patients with Type 2 Diabetes: A Systematic Review and Meta-Analysis

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ABSTRACT

Background: Coenzyme Q10 (CoQ10) is a naturally occurring antioxidant that has been suggested to have beneficial effects on lipid profiles and blood pressure. This systematic review and meta-analysis aim to evaluate the effects of CoQ10 supplementation on these parameters in patients with Type 2 Diabetes (T2D).
Objective: To assess the impact of CoQ10 supplementation on lipid profiles and blood pressure in individuals diagnosed with Type 2 Diabetes.
Methods: A systematic literature search was conducted in databases such as PubMed, Cochrane Library, and Scopus for randomized controlled trials (RCTs) published up to July 2024. Studies included were those that examined the effects of CoQ10 supplementation on lipid profiles (total cholesterol, LDL, HDL, triglycerides) and blood pressure (systolic and diastolic) in T2D patients.
Results: 16 studies were included. CoQ10supplementation reduced SBP (WMD: −3.86 mmHg, 95% CI: −6.01 to −1.71, $P = 0.014$, $I^2 = 83.7\%$; $P < 0.001$) and DBP (WMD: −2.70 mmHg, 95% CI: −4.50 to −0.91, $P = 0.024$, $I^2 = 92.1\%$; $P < 0.001$), but did not change lipid profile. Additionally, subgroup analysis indicated that the effects of CoQ10 on lipid profiles levels were more pronounced in studies where the daily dosage of CoQ10 was 100 mg or less, and the duration of the study was under 12 weeks.
Conclusions: Coenzyme Q10 supplementation appears to have a beneficial effect on lipid profiles and may contribute to lowering blood pressure in patients with Type 2 Diabetes. These findings suggest that CoQ10 could be a valuable adjunctive therapy for managing cardiovascular risk in this population. Additional in-depth research is needed to validate these findings and understand the underlying mechanisms in more detail.

Introduction

Diabetes mellitus is a metabolic disorder characterized by a lack of insulin or resistance to it, leading to high levels of glucose in the bloodstream (hyperglycemia).[1] The number of individuals affected by this condition is projected to rise to 439 million by 2030 and 642 million by 2040.[2] The metabolic irregularities and resulting complications of Type 2 Diabetes Mellitus (T2DM) have a substantial influence on the health and quality of life of patients.[3] These complications can include microvascular issues like retinopathy and nephropathy, as well

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as macrovascular problems affecting large blood vessels and peripheral vascular health.[4] Dyslipidemia, a common feature of Type 2 Diabetes, is characterized by elevated triglyceride and low-density lipoprotein cholesterol levels, along with reduced high-density lipoprotein cholesterol levels.[5,6] Treatment for T2DM involves a range of options such as insulin, alpha-glucosidase inhibitors, dipeptidyl peptidase 4 inhibitors, incretin analogs, insulin sensitisers, and intestinal lipase inhibitors.[7] These medications work to manage blood sugar levels and tackle the underlying metabolic abnormalities linked with the disease.[8–10] However, some of these therapies may have unwanted effects like gastrointestinal problems and weight gain.[11] Consequently, ongoing research efforts are focused on the development of new medications and natural compounds to improve the prevention and treatment of diabetes.[12] Coenzyme Q10 is a lipid-soluble compound found in cells that serves as an antioxidant, protecting cells from damage caused by free radicals.[13,14] Studies have shown that individuals with type 2 diabetes have lower levels of CoQ10 compared to those without diabetes. Some research has suggested that supplementing with CoQ10 may have positive effects on metabolic and hormonal markers in diabetes patients.[13,14] One study, for example, gave diabetic patients 120 mg of CoQ10 daily for 12 weeks and found significant reductions in triglycerides and VLDL levels, as well as improved insulin sensitivity.[15] While there have been several randomized clinical trials investigating the effects of CoQ10 supplementation on metabolic markers, there has yet to be a systematic review to determine its overall impact. Several randomized clinical trials (RCTs) have investigated the effects of CoQ10 supplementation on Metabolic Indicators, but no systematic review and meta-analysis has been conducted to define the effects of CoQ10 on Metabolic Indicators. In this review and meta-analysis, we collect details of randomized controlled trials to appraise the effectiveness of CoQ10 as a first agent in preventing oxidative stress.

Method

This study used a systematic review and meta-analysis following the design criteria specified in the PRISMA guidelines. The review process for this study is appropriately documented in the PROSPERO reviews repository; registration number CRD42024583103.

Search Strategy

The search strategy for this study was based on PICOS criteria. An unrestricted search was conducted in various international databases, including Embase, Web of Science, PubMed, Scopus, Google Scholar, and the Cochrane Library, up to May 2024, to identify relevant studies. The search terms used included both non-MESH and MESH-related topics. These terms were "Coenzyme Q10, [MESH]" "Q10, [MESH]" "ubiquinone [ti/ab]," "ubidecarenone [ti/ab]" or "ubiquinol, [ti/ab]" or "Bio-Quinone, [ti/ab]" or"diabetes, [MESH]" or "T2D, [MESH]" or "type 2 diabetes, [MESH]" or "diabetes, [MESH]" or "T2DM [MESH]" or "glycemic index, [ti/ab]" or "fasting blood glucose, [MESH]" or "insulin, [ti/ab]" or "glycated hemoglobin, [ti/ab]" or "glycemic indices [ti/ab]" or " GI [ti/ab]" and "DBP [ti/ab]" or "SBP [ti/ab]" or " systolic blood pressure, [MESH]" or " Dystolic blood pressure, [MESH]" AND lipid profile, [MESH]" or "high-density lipoprotein, [MESH]" or "HDL, [MESH]" or ""low-density lipoprotein, [MESH]" or VLDL, [MESH]" or "LDL, [MESH]" or "High-density lipoprotein, [MESH]" or "triglyceride, [MESH]" or "cholesterol, [MESH]" or "TG, [MESH]" or "TC [MESH]." We also have a list of carefully checked documents to ensure that no relevant studies are excluded from our review.

Inclusion and Exclusion Criteria

This review encompasses primary research studies that aligned with the specified requirements. The requirements involved: 1) individuals aged 18 years and above; 2) randomized controlled trials (RCTs) utilizing a placebo and following either a parallel or crossover design; 3) presentation of the average and standard deviation (SD) of MetS components at the commencement and conclusion of the trial for both the control and intervention groups; 4) trials that administered Q10 exclusively, without any other supplements or compounds. 5) Reporting the changes in and HDL-C, TG, LDL-C, TC, BMI, SBP and DBP during the intervention or at the beginning and end of it Studies that were not included in this review were those that: 1) were conducted on animals, in a lab setting, or in vivo; 2) were comprised of conference presentations, comments, observational research, letters, unpublished data, books, reviews, or meta-analyses; 3) centered solely on pregnant women and children; 4) did not allow for a comprehensive evaluation of the overall effects of Q10 supplementation. To ensure the accuracy of the selection process, two evaluators independently scrutinized the inclusion and exclusion criteria by thoroughly examining the titles, abstracts, and full texts of the studies. In instances of disagreement, a third evaluator participated in resolving the matter through deliberation.

Data Extraction

Various details were extracted from each study, including the main author, sample size, year of publication, trial location, study design, duration of coenzyme Q10 supplementation, and length of intervention. The data was meticulously managed by two authors to ensure accuracy. Any discrepancies that arose were resolved through collaborative discussions to ensure mutual agreement.

Risk of Bias Assessment

A risk assessment was conducted for each review using criteria developed by the Cochrane Collaboration. This approach encompasses various factors, including selection bias, reporting bias, perception bias, performance bias, and potential bias. The quality of evidence for each endpoint tested in each trial was categorized as low risk, high risk, or unknown risk of bias. Each assessment of bias is based on whether there is a high risk of bias in one or more of the major factors, a low risk of bias in all factors, or uncertainty regarding bias in all factors. In the latest meta-analysis, each study was evaluated for its quality based on whether it was rated as "low risk" across all criteria or as "high risk" for at least one criterion according to the Cochrane Risk of Bias Assessment Tool (Table 1). Any discrepancies in the assessment scores were resolved by consulting the third author.

Statistical Analysis

We performed statistical analysis using Stata software version 16.0 developed by StataCorp LP in College Station, Texas, USA. Statistical significance was determined by a *P*-value less than 0.05. The formula utilized to compute the standard deviation (SD) of the mean difference was: SD = $\sqrt{}$(SD pretreatment)ˆ2 + (SD release-treatment)ˆ2 - (2R SD pretreatment SD release correction), where R equals 0.5 [27] (a). SD was calculated based on the SEM × $\sqrt{n}$ only if the SEM became statistically significant, with n denoting the population size. The mean and standard deviation of the variables were employed to derive the overall outcomes. The variables' dimensions were presented as the weighted mean difference (WMD) with a 95% confidence interval (CI) in the C programming language. The I2 index gauged the analysis's heterogeneity, with I^2>50% indicating substantial heterogeneity. The results were leveraged for a meta-analysis concerning variable distinctions; otherwise, a model with constant values was utilized. A sensitivity analysis was conducted to assess each appropriate test's impact on the comprehensive estimation. Begg and Egger conducted an experiment to evaluate e-book bias through visual analysis of funnel plots. The non-linear impact of intervention duration (weeks) and Coenzyme Q10 dose (mg/day) was evaluated using a prevalent model.

Table 1
Study characteristics.

First author	Year/ Country	Study design	Subject	Participants (intervention/ control)	Mean age (Intervention/ Control)	Baseline BMI (Intervention/ Control)	Duration (wk)	dose	Type of administration		Results
									Intervention	Placebo	
Akbari Fakhrabadi, M et al.	2015/ Iran	RCT parallel	Diabetic Neuropathy	32/ 30	56.7/ 54.8	28.7/ 29.6	12	200 mg/d	coenzyme Q10	placebo	Significant change: insulin sensitivity HOMA-IR, Unchanged: BMI
Andersen C B et al	Denmark/ 1997	RCT parallel	Patients with Insulin Dependent Diabetes Mellitus	34 (17/ 17)	35/ 35.3	23.5/ 24	12	100 mg/d	capsule coenzyme Q10	placebo	Unchanged: HDL, LDL,BMI
Chew G T et al	Australia/ 2008	RCT parallel	T2DM With Left Ventricular Diastolic Dysfunction	36 (16/20)	61.3/ 62.4	30.1/ 30.7	24	200 mg/d	coenzyme Q10	placebo	Unchanged: HDL, LDL, TC, TG, FBS, SBP, DBP, BMI
Fallah M et al.	2018/ Iran	RCT parallel	diabetic hemodialysis patients	60 (30/30)	59.4/ 64.8	26.9/ 26.6	12	120 mg/d	coenzyme Q10	Placebo	Unchanged: HbA1c, BMI
Eriksson J G et al	Finland/ 1999	RCT parallel	T2DM	23 (12/ 11)	65/ 64	29/ 29.8	24	100 mg/d	capsule coenzyme Q10	placebo	Unchanged: HDL, TC, TG, FBS, HbA1c, SBP, DBP, BMI
Gholami M et al.	Iran/ 2019	RCT parallel	T2DM	70 (35/35)	52.97/ 53.68	29.3/ 28.51	12	100 mg/d	coenzyme Q10	placebo	Significant change: FBS, HOMA-IR, TC, LDL-C, FBS Unchanged: SBP, DBP,TG BMI, HbA1c
Hamilton S J et al.	Australia/ 2009	RCT crossover	Statin-Treated T2DM	23 (23/23)	68/ 68	29/ 29	12	200 mg/d	coenzyme Q10	placebo	Unchanged: SBP, DBP,LDL
Hernández-Ojeda J et al	Mexico/ 2012	RCT parallel	diabetic polyneuropathy	49 (24/25)	55.3/ 57	29.4/ 29.3	12	400 mg/d	ubiquinone	placebo	Unchanged: HDL, TC, TG, FBS, HbA1c, SBP, DBP
Hodgson J. M et al	Australia/ 2002	RCT parallel	T2DM	37 (19/ 18)	52.3/55.2		12	200 mg/d	oral coenzyme Q10	placebo	Significant change: SBP, DBP Unchanged: TG BMI, HbA1c, FBS, TC, LDL-C,
Hosseinzadeh Attar M J et al.	Iran/ 2015	RCT parallel	T2DM	64 (31/33)	45.2/ 47.1	29.52/ 29.47	12	200 mg/d	oral coenzyme Q10	placebo	Unchanged: FPG, HbA1c, TC, TG, LDL-C and HDL-C, BMI
Mehrdadi P et al	Iran/ 2017	RCT parallel	T2DM	56 (26/30)	46/ 48	29.68/ 29.31	12	200 mg/d	coenzyme Q10	placebo	Unchanged: FBS, HOMA-IR, Insulin, HbA1c,
Moazen M et al	Iran/ 2015	RCT parallel	T2DM	52 (26/26)	50.67/ 52.79	25.31/ 25.34	8	200 mg/d	coenzyme Q10	placebo	Unchanged: FBS
Mohammed-Jawad N K et al.	Iraq/ 2014	RCT parallel	T2DM	38 (19/19)	49.37/ 51.63	28.15/ 29.5	8	150 mg/d	coenzyme Q10	control	Significant change: HbA1c, FBS, TC, LDL-C Unchanged: HDL,BMI
Rodríguez-Carrizalez A D et al.	Mexico/ 2016	RCT parallel	nonproliferative diabetic retinopathy	40 (20/20)	58.5/ 57.8	28.2/ 29.3	24	400 mg/d	coenzyme Q10	placebo	Unchanged: TG, Glucose, LDL-C, HDL, TC, BMI
Yen C H et al.	Taiwan/ 2018	RCT parallel	T2DM	47 (24/23)	61.5/ 59.6	28/ 27.3	12	100 mg/d	liquid ubiquinol	placebo	HbA1c,HDL Unchanged: TG, BMI, FBS, TC, LDL-C, FBS, HOMA-IR, Insulin
Zahedi H et al.	Iran/ 2014	RCT parallel	T2DM	40 (20/20)	53.5/ 58.8		12	150 mg/d	coenzyme Q10	placebo	Significant change: FPG and HbA1C, TG, LDL-C, HDL Unchanged: HOMA-IR

RCT = randomized clinical trial. OPE = quercetin rich onion peal extract; BW = body weight; BMI = body mass index; WC = waist circumference; HC = hip circumference; WHR = waist-to-hip ratio; BF = body fat percent; TG = triglyceride; TC = total cholesterol; HDL = high-density-lipoprotein; LDL = low-density-lipoprotein; FBS = fasting blood glucose; Glu = glucose; Ins = insulin; HbA1c = hemoglobin A1C; SBP = systolic blood pressure; DBP = dyastolic blood; FPG = fasting plasma glucose.

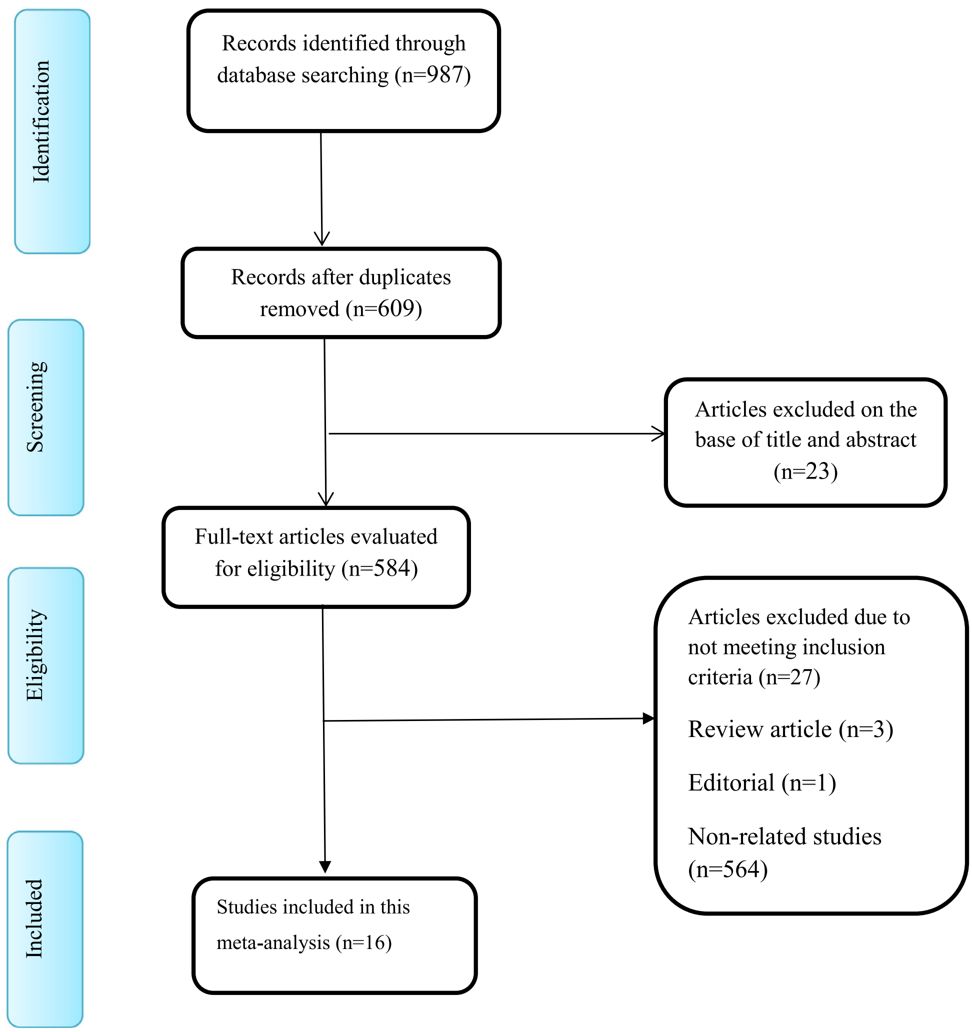

Figure 1. Study flowchart.

Results

Study Selection

A entire of 987 articles were obtained through database searches. Of those, 378 duplicate records were eliminated, leaving 609 articles for subsequent review of titles and abstracts. Following this review, another 584 articles were eliminated due to not meeting the inclusion criteria leaving a total of 25 potentially eligible papers. Upon retrieving full manuscripts of the 25 articles, an additional 9 papers were eliminated leaving a final total of 16 papers that met the inclusion criteria (Figure 1).[21–35]

Study Characteristics

The included studies were published between 1997, and 2019. The countries where studies were conducted included Denmark,[16] Australia,[17,21,23] Iran,[19,20,24–26,30,31] Iraq,[27] Finland,[18] Mexico,[22,28] and Taiwan.[29] Among the included studies, one was conducted on females,[20] and the remaining on both sexes. The mean age of the individuals was between 35, and 68 years, and their mean BMI was between 23.50, and 30.10 kg/m^2. The dose of CoQ10 supplementation varied from 100 to 400 mg/day, and the duration of supplementation was between 8 and 24 weeks. Table 1 provides a summary of the key aspects of the studies that were included. For a more in-depth analysis of the bias risk assessment within each category, please refer to Supplementary Table 2.

Impact of CoQ10 Supplementation on SBP

The combination of 10 various effect assessments demonstrated that including CoQ10 led to a significant decrease in serum SBP levels when compared to the control groups (WMD: -3.86 mmHg, 95% CI: -6.01 to -1.71, $P = 0.014$, $I^2 = 83.7\%$; $P < 0.001$) (Figure 2A). The subgroup analysis indicated a greater reduction effect of CoQ10 supplementation in the following scenarios: studies with more than 50 participants, study durations of less than 12 weeks, CoQ10 dosages of 100 mg/day or less, and among individuals aged 55 years or younger (Supplementary Table 3).

Impact of CoQ10 Supplementation on DBP

The synthesis of 10 different effect measurements demonstrated that adding CoQ10 to the diet resulted in a notable reduction in serum DBP levels in contrast to the control groups (WMD: -2.70 mmHg, 95% CI: -4.50 to -0.91, $P = 0.024$, $I^2 = 92.1\%$; $P < 0.001$) (Figure 2A). The subgroup analysis indicated a greater reduction effect of CoQ10 supplementation in the following scenarios: studies with more than 50 participants, study durations of less than 12 weeks, CoQ10 dosages of 100 mg/day or less, CoQ10 dosages of 100 mg/day or less, and among individuals aged 55 years or younger (Supplementary Table 3).

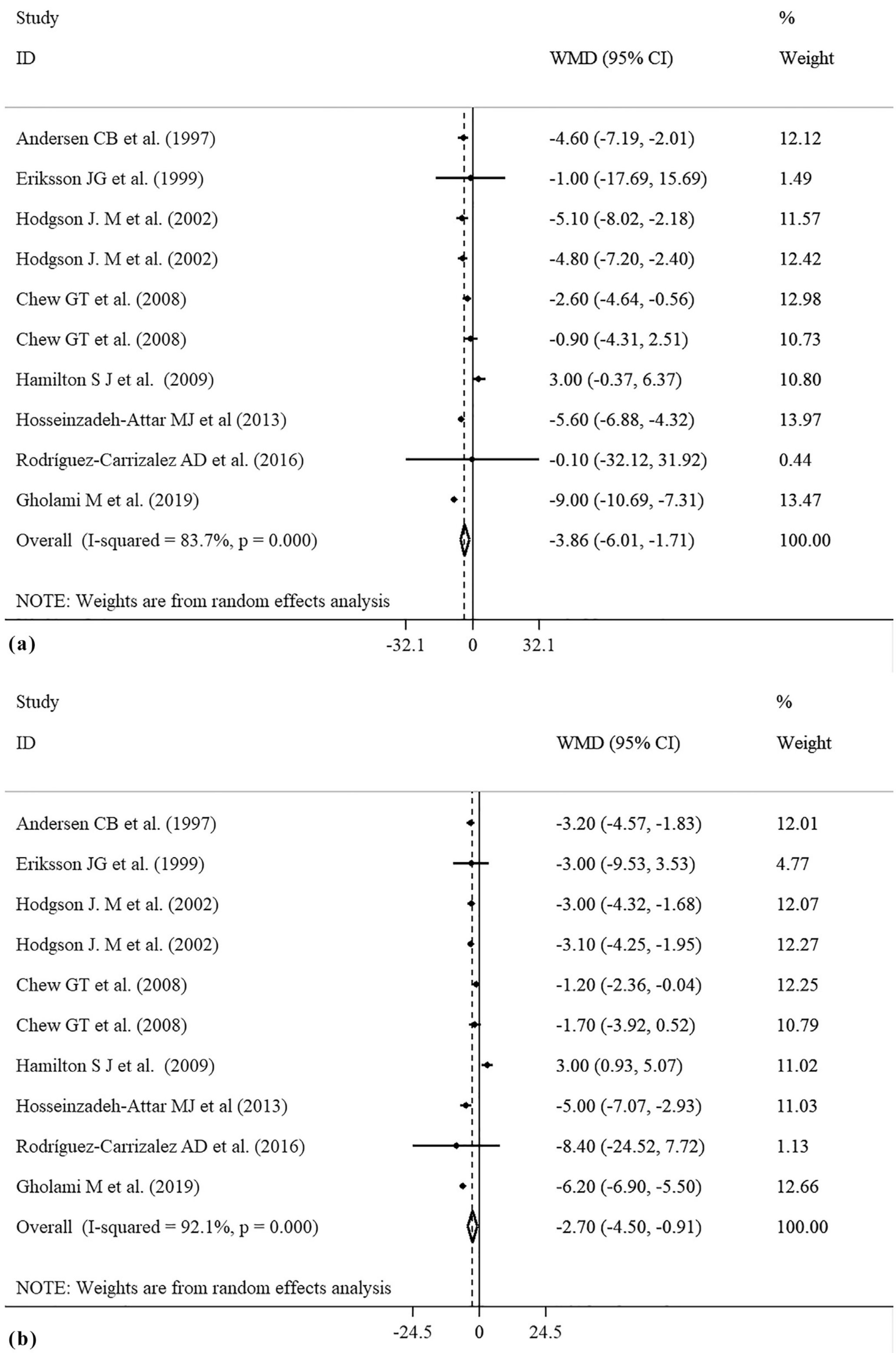

Figure 2. Blood pressure forest plot. a) systolic blood pressure forest plot; b) diastolic blood pressure forest plot.

Effect of CoQ10 Supplementation on TG

The analysis of 10 different effect sizes revealed that consumption of CoQ10 resulted in a slight reduction in serum triglyceride levels compared to the control groups, though this difference was not statistically significant (WMD: -14.14 mg/dL, 95% CI: -29.67 to 1.38, P = 0.074) (Figure 3A). Also, moderate heterogeneity among the included studies was detected (I^2 = 77.4%; P < 0.001). The subgroup analysis revealed sample size, duration, and dose were sources of heterogeneity. The subgroup analysis indicated a greater reduction effect of CoQ10 supplementation in the following scenarios: studies with more than 50 participants, CoQ10 dosages of 100 mg/day or less, study durations of less than 12 weeks, and among individuals aged 55 years or younger (Supplementary Table 3).

Impact of CoQ10 Supplementation on TC

A comprehensive review of 12 different studies showed that taking CoQ10 supplements did not have a noticeable impact on serum TC levels when compared to groups who did not take the supplements (WMD: -2.86 mg/dL; 95% CI: -10.59 to 4.88; p = 0.469) (Figure 3B). Additionally, there was a significant amount of variability among the studies included in the analysis (I^2 = 94.1%; P < 0.001). Further analysis into subgroups revealed that factors such as dosage played a role in the heterogeneity observed. The subgroup analysis indicated a greater reduction effect of CoQ10 supplementation in the following scenarios: studies with more than 50 participants, and CoQ10 dosages of 100 mg/day or less (Supplementary Table 3).

Impact of CoQ10 Supplementation on LDL-C

The combination of 13 studies showed a nonsignificant reduction of serum LDL-C levels following CoQ10 supplementation compared to the control groups (WMD: −0.80 mg/dL; 95%CI: −10.62 to 9.02; P = 0.873) (Figure 3C). Also, the heterogeneity among the included studies was

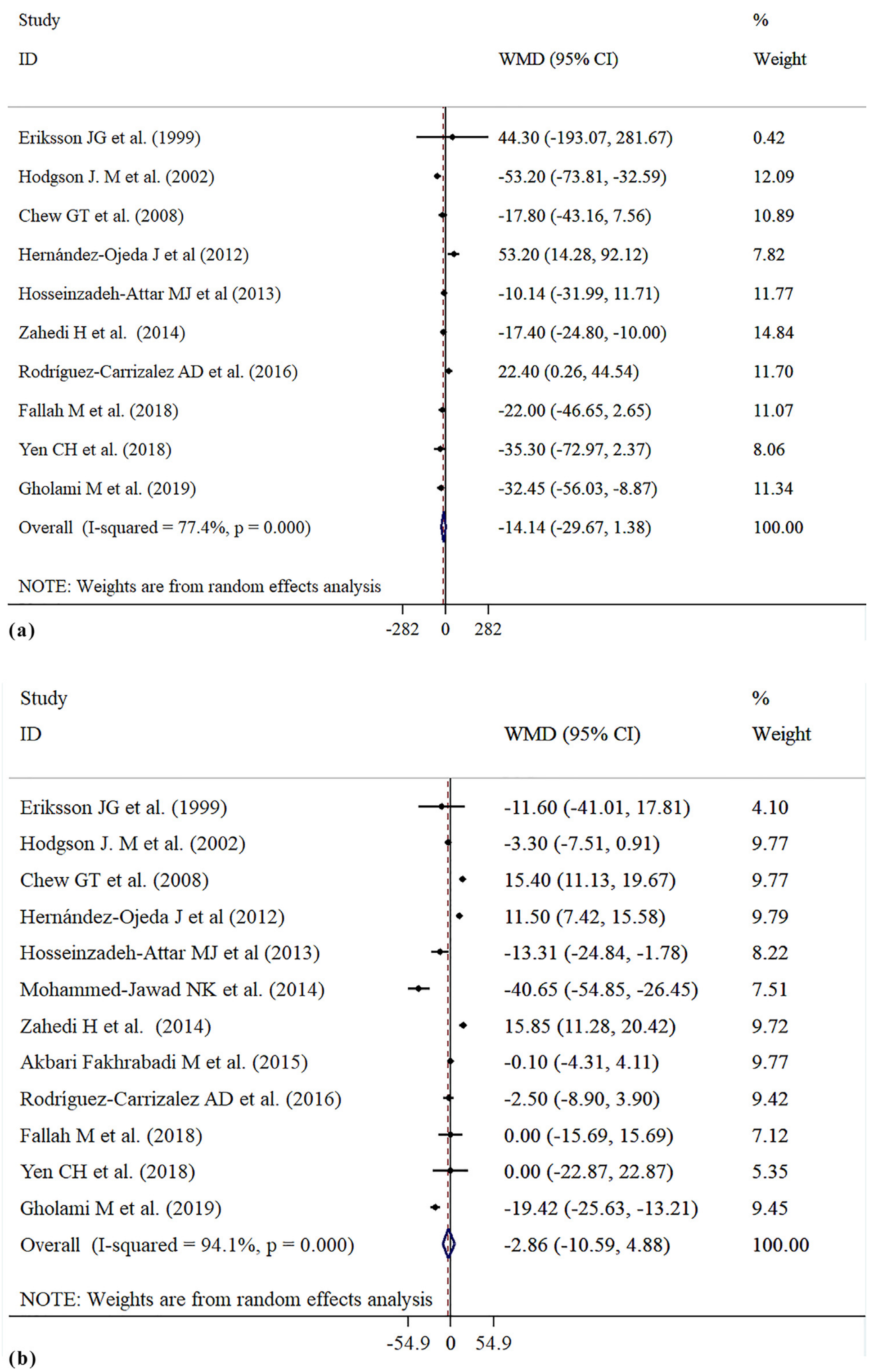

Figure 3. Lipid profile forest plot. a) TG forest plot; b) TC forest plot; c) LDL-C forest plot; d) HDL-C forest plot.

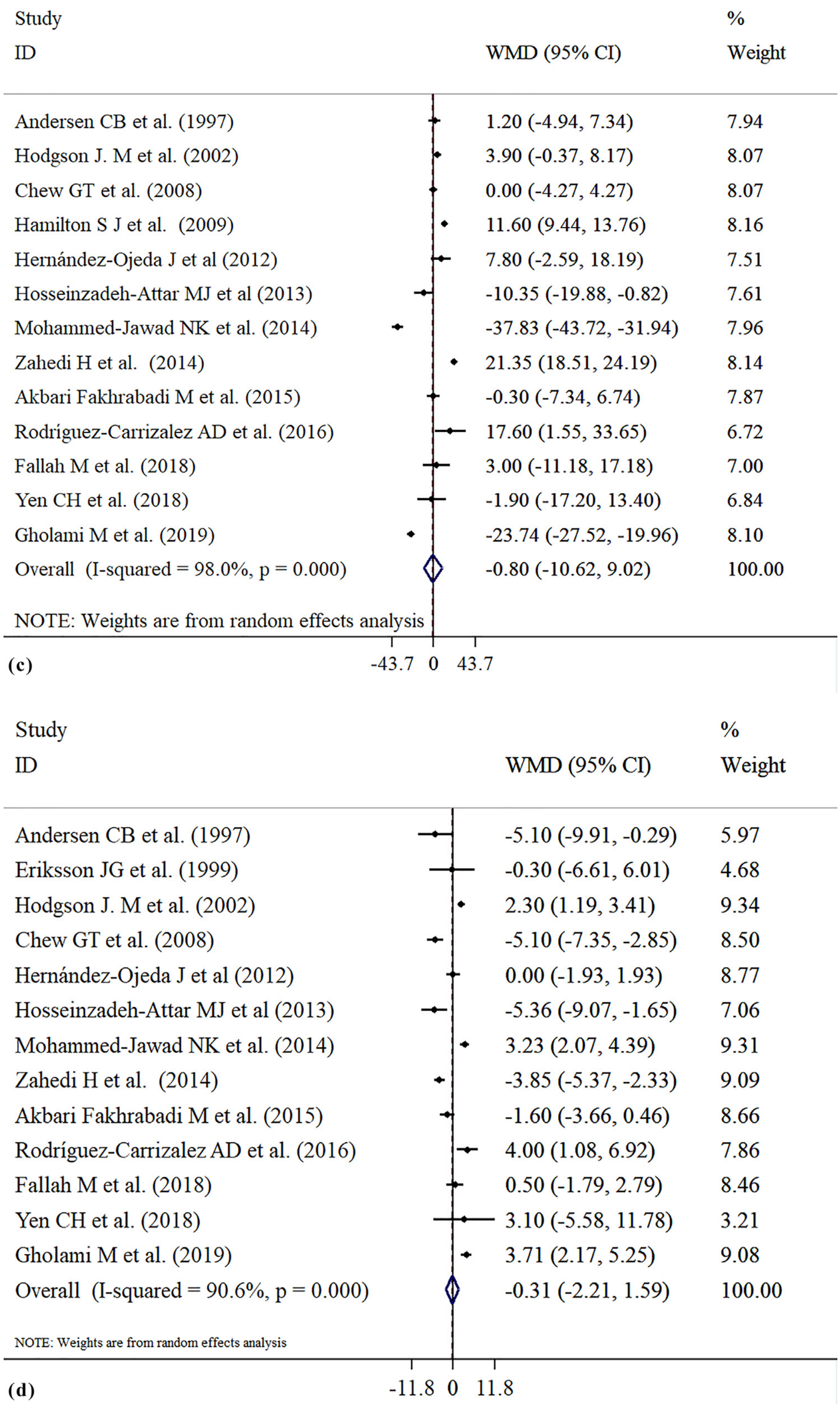

Figure 3. Continued

significant (I^2 = 98%; P < 0.001). The subgroup analysis indicated a greater reduction effect of CoQ10 supplementation in the following scenarios: studies with more than 50 participants, and CoQ10 dosages of 100 mg/day or less (Supplemental Table 3).

Impact of CoQ10 Supplementation on HDL-C

Combining data from 13 studies, it was found that CoQ10 supplementation did not have a significant impact on HDL-C levels when compared to control groups (WMD: −0.30 mg/dL; 95% CI: −2.21 to 1.59; P = 0.749) (Figure 3D). Additionally, there was a significant variation between the studies included (I^2 = 90.6%; P < 0.001). Subgroup analysis indicated that CoQ10 supplementation had a more pronounced increase effect in studies with over 50 participants, CoQ10 dosages of 100 mg/day or less, and among individuals aged 55 years or younger (Refer to Supplemental Table 3).

Sensitivity Analysis and Publication Bias

After omitting several studies, results of the sensitivity analysis showed no appreciable relationship between CoQ10 supplementation and SBP, DBP, TC, LDL-C, HDLC, and insulin. In order to evaluate publication bias, funnel plots along with Egger's and Begg's tests were employed. Upon visual examination, asymmetry was detected in the funnel plots. However, the results of both Begg's and Egger's tests indicated no indication of publication bias for SBP (P = 0.765, 0.376), DBP (P = 0.721, 0.341), TG (P = 0.474, 0.650), TC (P = 0.837, 0.145), LDL-C (P = 0.760, 0.258), and HDL-C (P = 0.583, 0.204) (Supplementary Figure 1 a-f).

Discussion

This systematic review and meta-analysis included information from 16 relevant studies that investigated the effects of Coenzyme Q10 (CoQ10) on lipid levels and blood pressure in people with T2DM. Our analysis demonstrated that supplementation with CoQ10 led to notable decreases in SBP, and DBP. Additionally, subgroup analysis indicated that the effects of CoQ10 on lipid profiles levels were more pronounced in studies where the daily dosage of CoQ10 was 100mg or less, and the duration of the study was under 12 weeks. Recent research has pointed out that oxidative stress contributes to the onset of high blood sugar, insulin resistance, and impaired beta cell function. While the exact mechanism by which CoQ10 influences lipid profiles remains unclear, several theories have been proposed. CoQ10, being an intracellular antioxidant, shields the cell membrane phospholipids and mitochondrial membrane protein from damage caused by free radicals.[26] Aldehyde derivatives resulting from lipid peroxidation, such as malondialdehyde, can inhibit lecithin-cholesterol acyl transferase (LCAT), which is responsible for esterifying free cholesterol on HDL-C. Therefore, CoQ10 may enhance the production of HDL-C, curb oxidative stress, and consequently decrease malondialdehyde levels.[27] Furthermore, the intake of CoQ10 might stimulate the gene expression of peroxisome proliferator-activated receptor-γ (PPAR-γ) by activating the calcium-mediated AMPK pathway and inhibiting adipogenesis induced by differentiation.[28] PPAR-γ is a nuclear receptor protein that functions as a ligand-activated transcription factor, regulating gene expression that impacts insulin and lipid metabolism, differentiation, proliferation, survival, and inflammation.[29] Numerous theories have been proposed to explain how CoQ10 helps lower blood pressure. One of the key mechanisms is believed to be its antioxidant properties. Oxidative stress can diminish the availability of nitric oxide, which in turn causes blood vessels to constrict, leading to high blood pressure. However, CoQ10 supplementation can boost antioxidant levels and increase nitric oxide availability, directly benefiting the endothelium.[32,33] A study combining data from 5 randomized controlled trials involving 194 participants showed that CoQ10 improved endothelial function by enhancing flow-mediated dilatation.[32,35] Another way CoQ10 may reduce blood pressure is by mimicking the effects of angiotensin, thus promoting excretion of sodium and lowering levels of aldosterone, a hormone that affects blood pressure.[33,36] Furthermore, Coenzyme Q10 (CoQ10) has been shown to have a positive impact on cardiovascular health by stimulating the production of prostacyclin, a potent vasodilator. This compound also enhances the responsiveness of arterial smooth muscles to prostacyclin, which promotes healthy blood flow.[34,37] Additionally, the hypoglycemic and hypolipidemic properties of CoQ10 supplementation can help lower blood pressure.[37] Elevated levels of blood sugar and lipids are known to impair endothelial function and increase the risk of developing cardiometabolic disorders.

Limitations and Strengths

This meta-analysis has a number of limitations and strengths that are worth discussing. The current study is a thorough, up-to-date systematic review and meta-analysis of previous research on the impact of CoQ10 on metabolic indicators in patients with T2DM. It is also notable that participants in the included RCTs included various health statuses which may have impacted the generalizability of study outcomes. Apart from the aforementioned limitations, this meta-analysis has some important strength including the fact that RCTs meeting the inclusion criteria were not limited in publication date or language in order to generate the most comprehensive systematic review on CoQ10 to date. As a result, included studies were undertaken in a variety of countries on different continents, thus increasing the generalizability of results. A standardized methodology and robust statistical methods were utilized to assess the effects of CoQ10 on T2DM. Since heterogeneity was apparent, subgroup analyses further elucidated on outcomes related to CoQ10 supplementation.

Conclusions

Our findings suggest that CoQ10 supplementation may be potentially effective for clinically reducing SBP, DBP and TG in patients with T2DM. To further determine the exact recommended intake of CoQ10 for lowering blood pressure and lipid profile, we are also designing a clinical trial with different doses of CoQ10 intervention and are preparing to carry out this study in the near future.

Data availability Statement

The data that support the findings of this study are available on request from the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Yinshuang Li: Methodology, Investigation. **Amr Ali Mohamed Abdelgawwad El-Sehrawy:** Conceptualization, Resources, Data curation, Visualization, Supervision, Project administration, Funding acquisition. **Amar Shankar:** Software. **Manish Srivastava:** Validation. **Jaafaru Sani Mohammed:** Formal analysis. **Ahmed Hjazi:** Investigation, Writing – original draft. **I.B. Sapaev:** Writing – original draft, Writing – review & editing. **Yasser Fakri Mustafa:** Writing – original draft, Writing – review & editing. **Munther Kadhim Abosaoda:** Writing – review & editing.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.clinthera.2024.12.010.

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